

Comparing Classical Machine Learning Classifiers for Frame-Level Recognition of Mouse Social Behaviour from Pose-Tracking Data

SYDE/BME 522: Foundations of Artificial Intelligence, Fall 2026 | Instructor: Dr. Charbel Azzi

Group 24 | Alex Rus, Daniel Wu, Kaviya Gopalakrishnan, Wallace Lee

Project Option 2: Empirical Evaluation

Behavioural neuroscience labs routinely film pairs of mice to study social behaviour, but converting that footage into usable data still relies on experts labelling every frame by hand as *attack*, *investigation*, *mount*, or *other*. This is slow, taking hours of annotation per video, and inconsistent between annotators, which limits both throughput and reproducibility [1, 2]. At the same time, pose-estimation and tracking systems have made it inexpensive to extract body-part coordinates for each animal, so the bottleneck has shifted from *tracking* mice to *interpreting* their tracks. Automated behaviour classification is therefore a relevant, practical problem: a reliable classifier would save labs substantial annotation time and apply one consistent standard across experiments. Our project follows **Option 2 (Empirical Evaluation)**: we will implement and benchmark several established supervised learning techniques to determine how well expert behaviour labels can be recovered from pose data alone, and which design choices matter most. We will use Task 1 (Classic Classification) of the Caltech Mouse Social Interactions (CalMS21) dataset [1], which provides top-down recordings of a two-mouse resident–intruder assay already converted into seven keypoints per mouse per frame (nose, ears, neck, hips, and tail base), each frame carrying an expert label from the four classes above.

This is a supervised multi-class classification problem. We will first convert the raw keypoint coordinates into features that describe how the two mice relate to each other, such as the distance between them, their speeds, and their relative orientation, since behaviour depends on these relationships rather than on where the mice are in the cage. To measure the value of temporal information, we will compare models using spatial features alone with models using spatial and motion features to see how short-term temporal information supports classification. We will then train and compare several standard classifiers on these features, including **k-nearest neighbours**, a **random forest**, a **support vector machine**, and a small **neural network**, evaluating each on videos held out from training. Because most frames are labelled *other*, accuracy alone would be misleading, so we will report F1 scores for each behaviour. We also plan to examine which behaviours are most often confused with one another and which features are most informative. Published results from the CalMS21 baselines [1] and the MARS pipeline [2] will serve as reference points for our findings.

References

- [1] J. J. Sun, T. Karigo, D. Chakraborty, S. P. Mohanty, B. Wild, Q. Sun, C. Chen, D. J. Anderson, P. Perona, Y. Yue, and A. Kennedy, “The Multi-Agent Behavior Dataset: Mouse Dyadic Social Interactions,” in *Proc. Neural Information Processing Systems (NeurIPS) Track on Datasets and Benchmarks*, 2021.
- [2] C. Segalin, J. Williams, T. Karigo, M. Hui, M. Zelikowsky, J. J. Sun, P. Perona, D. J. Anderson, and A. Kennedy, “The Mouse Action Recognition System (MARS) software pipeline for automated analysis of social behaviors in mice,” *eLife*, vol. 10, e63720, 2021.

Dataset: CalMS21 Task 1 (Classic Classification):

<https://data.caltech.edu/records/s0vdx-0k302>

Download: `task1_classic_classification.zip`.